

ARMY CONTRACTING COMMAND, ABERDEEN PROVING GROUND
EDGEWOOD CONTRACTING DIVISION
W911SR-25-S-CSIRP
ANNOUNCEMENT OF CSIRP INDUSTRY DAY:

The Army Contracting Command–Aberdeen Proving Ground (ACC-APG) Contracting Center, Edgewood Contracting Division, Ft Detrick, in support of the Joint Program Executive Office for Chemical, Biological, Radiological, and Nuclear Defense (JPEO CBRND), Joint Program Manager, CBRN Sensors (JPM CBRN Sensors), hereby issues an announcement in advance of a Joint Industry Day for CBRN Sensor Integration on Robotic Platforms (CSIRP) effort to be held virtually via MS Teams on **21 May 2025**. This is not a solicitation, nor is this a solicitation requesting proposals at this time.

The purpose of this Industry Day is to identify industry partners and capabilities to develop/deliver the engineering and integration of payloads with ability to detect and identify additional threats and perform sample collection for use on unmanned air, ground, and water-borne surface platforms. These payloads should transmit CBRN data through a common operating picture giving warfighters the ability to integrate CBRN expertise and knowledge into all echelons, providing situational awareness in mounted and dismounted configurations enabling decisions in austere environments. Appendix A outlines the draft Statement of Objectives (SOO).

This Industry Day will include a presentation by JPM CBRN Sensors as well as opportunities for industry partners to present their capabilities in one-on-one breakout sessions with representatives from the Government agencies. Industry partners may update Government stakeholders, partners, and end users on current efforts and capabilities.

Companies that wish to attend must complete the attached registration form and return to the identified POC's no later than COB **15 May 2025**. White Papers will need to be submitted by COB **19 May 2025**. Please limit attendees to no more than 5 and confirm your interest in presenting your capabilities to the Government either in a group forum, or separately one on one. Those requesting to register must also provide a non-binding 3–5-page White Paper demonstrating their company's capability encompassing the following:

- Company overview
- Capability to meet the requirements of Appendix A.
- Questions or comments in advance.
- Anticipated ROM.

Proprietary information should be identified as Company Proprietary. Do not use Government security classification markings.

This Industry Day notification is released under announcement number W911SR-25-S-CSIRP. The Government is not liable to reimburse any costs incurred by companies or entities that respond in any way to this announcement or participate in the Industry Day. This is industry day is for planning purposes only. It is not to be construed as a commitment by the Government nor will the Government pay for the information solicited. No solicitation document exists or is guaranteed to be issued as a result of this announcement or industry day. There is no Draft Request For Proposal that has been issued.

Additional details on location, times, agenda etc. will be shared with registered participants only.

APPENDIX A

ATTACHMENT 1 STATEMENT OF OBJECTIVES

PROJECT TITLE: T CBRN Sensor Integration on Robotic Platforms (CSIRP) Capability Set 4.0 – Expanded CBRN Detection and Autonomy

BACKGROUND: The Joint Program Executive Office for Chemical, Biological, Radiological, and Nuclear Defense (JPEO-CBRND), Joint Project Manager for Chemical, Biological, Radiological and Nuclear (CBRN) Sensors (JPM CBRN Sensors) and Chemical, Biological, Radiological, and Nuclear (CBRN) Sensor Integration on Robotics Platforms (CSIRP) seeks the engineering and integration of payloads with ability to detect and identify additional threats and perform sample collection for use on unmanned air, ground, and water-borne surface platforms. These payloads should transmit CBRN data through a common operating picture giving warfighters the ability to integrate CBRN expertise and knowledge into all echelons, providing situational awareness in mounted and dismounted configurations enabling decisions in austere environments.

The objective of this prototyping effort is to further explore the employment of current and future CBRN sensors on service-relevant unmanned aircraft system (UAS), unmanned ground system (UGS), and unmanned surface vessel (USV) platforms across several missions prioritized by the Department of Defense user community. These efforts are focused on demonstrating integrated CBRN robotics capabilities to inform requirements and concepts of operation (CONOPS) and understand the capabilities and limitations of the completed efforts.

PROJECT OBJECTIVES: This project intends to facilitate the integration of CBRN sensors on unmanned systems to enhance the ability to detect and identify additional threats through a common operating picture and to perform sample collection when needed in CBRN contested environments. The SOO covers the engineering and repackaging, integration (hardware and software), and testing to support future employment. Service input was provided from the following:

United States Army (USA); Maneuver Support Center of Excellence (MSCOE); National Guard Bureau (NGB); United States Marine Corps (USMC); United States Navy (USN); United States Special Operations Command (USSOCOM); and United States Air Force (USAF) Explosive Ordnance Disposal (EOD) and Emergency Management/CBRN.

Objective 1: Biological Point Detection/Identification Payloads: Most biological detection relies on particle sizing and sample collection followed by sample processing in a lab for confirmation. New advances in biological hazard detection and identification technology have rendered those technologies smaller, lighter, and more viable on an unmanned platform. CSIRP seeks to integrate mature sensors to detect and identify potential biological hazards as well as collect samples for confirmatory analysis.

- Produce a more accurate biological detection and identification sensor that can do sample collection and provide real time confirmation of hazard in the battlefield.
- Integrate novel reduced-size, weight, and power (SWAP) Biological detection and identification system onto UASs, UGSs, and USVs.

Objective 2: Microsensor Payloads: Soldiers will deploy CBRN Microsensors (C-MS) throughout the operating environment from a remote and/or stand-off location to provide continuous detection and identification of CBRN threats.

- CSIRP is looking to integrate C-MS payloads and employment mechanisms onto various UAS and UGS.
- Achieve integrated layered defense through deployment of remote sensing solutions for CBRN detection and identification.
- Provide real time understanding of CBRN threats by integrating into current Army and Joint networks and command and control systems.

Objective 3: Expanded Detection and Autonomy Capabilities: Next generation handheld chemical detection and identification will be possible with the Compact Vapor Chemical Agent Detector (CVCAD) and the Proximate Chemical Agent Detector Non-Trace (PCAD NT).

- Replace legacy chemical sensors with more advanced and reduced SWAP sensing options on UGS and UAS platforms. Interfaces should be platform agnostic to allow for use on newer platforms without extensive adaptation.
- Integrate software for more advanced sensing solutions into a common command and control operating picture.

Objective 4: Command and Control Integration: The ability to make informed, fast decisions on the battlefield is critical for warfighters' maneuverability, operations and survivability. CSIRP seeks to have sensors data integrated into communication networks that create decision support tools that provide the warfighter with information at their fingertips. CSIRP seeks to seamlessly incorporate various sensor technologies into one operating picture. Systems and capabilities communicating with each other and working together to provide situational understanding will get actionable information to the joint force faster.

- Integrate current chemical and radiological sensors along with a biological identification sensor, CVCAD, and PCAD NT, into an automated CBRN hazard warning and reporting common user interface and unified software solution.

Interfaces should be platform agnostic to allow for use on newer platforms without extensive adaptation.

- Provide integrated situational awareness on potential CBRN hazards to inform decision making with next generation warning reporting and advanced hazard prediction and decision support tools.
 - Give access to CBRN data and information collected during a mission to be shared in one holistic picture through a set of networked sensors.
 - Ensure CSIRP integrated sensors have the ability to be plugged into service specific computing environments providing continuous CBRN defense capability to the joint force through a unified software solution.
 - Provide CBRN layered defense through integration of newer sensor technologies, real time classification of data through common interfaces and data sharing displays.
- Interfaces should be platform agnostic to allow for use on newer platforms without extensive adaptation.

DATA DELIVERABLES: The Government intends to manage development towards the above-listed objectives using the following deliverables. The Awardee(s) shall submit deliverables using a format in accordance with the OTA Base Agreement and/or as negotiated with the Agreement Officer's Representative (AOR).

Technical data deliverables described herein shall be delivered to the Government with Government Purpose Rights.

1 Integrated Master Schedule (IMS): The Government expects to manage the program using a shared integrated master schedule. The member should propose the creation, maintenance, update of an IMS as well as execution based upon that master delivery schedule. The IMS will be shared with the Government program team. It is expected that the IMS will detail the durations of tasks/sub-tasks. The Government expects that the IMS will be updated no less frequently than on a monthly basis. The statement of work (SOW) should address this requirement fully.

2 Preliminary Design Review (PDR): The Awardee(s) will prepare and deliver a briefing sufficient to address design trade-off and a risk assessment for the Course of Action (COA) to complete the outlined task(s).

3 Software Design Lifecycle Documentation (SDLD): The Awardee(s) will deliver SDLD using the Defense Industrial Base and Department of the Army Best Practices. The Awardee will prepare and submit Software Development Package (SDP) that includes all interface and data files.

4 Risk Management Framework (RMF) Documentation: The Awardee(s) will deliver RMF using the Defense Industrial Base and Department of the Army Best Practices. The Awardee will prepare and submit a System Security Plan that includes all baseline controls of

a system categorized at the confidentiality level: Moderate, integrity level: Moderate, and availability level: Moderate in addition to any controls required by the appropriate Security Technical Implementation Guides (STIGs).

5 Critical Design Review (CDR): The Awardee(s) will prepare briefing materials and documents to conduct a CDR that details the systems design, evaluated/anticipated performance, cybersecurity findings, manufacture approach, sustainment, safety, and environmental factors.

6 Configuration Control: After CDR, the offeror will use standard business practice Configuration Control processes. In addition, after the Technical Data Package (TDP) is delivered, the offeror will use Design Change Logs to document changes to the TDP. The Design Change Logs, which could include TDP redlines or markups, will clearly document the change's purpose and potential benefit/impact to system performance, test scope, cost, and schedule.

7 Technical Data Package (TDP): The Awardee(s) will deliver a TDP that includes drawings, schematics, and software/firmware source code. The Awardee(s) will develop and deliver the initial TDP at CDR and upon delivery of the manufactured prototypes. The final TDP will include all changes stemming from Developmental Testing (DT) and submitted with the Final Report

8 Safety Assessment and Environmental Report: The Awardee(s) will document any potential safety and health hazards, safety and health risks (i.e., severity and probability of occurrence for each identified hazard), hazardous materials (HAZMAT), and special procedures or precautions associated with the system's manufacture, operation, repair, storage, and disposal

9 Test Reports: The Awardee(s) will summarize and document results from all test events in a final report. The Awardee(s) will provide quick-look reports within 15 days after the conclusion of each test event, with the final report combining and expanding on each quick-look report within 60 days from the conclusion of the test event.

10 Bi-Weekly Status Meetings: Every other week the Awardee(s) will meet internally with JPM CBRN Sensors / CSIRP to discuss costs, technical progress, and schedule.

11 Quarterly Report: The Awardee(s) will prepare a Quarterly Report which will include a Technical Status Report and a Business Status Report in accordance with the terms and conditions of the Base Agreement. The Awardee(s) will be prepared to brief the JPM CBRN

Sensors, the Joint Product Manager Reconnaissance and Platform Integration (JPdM RPI), and external stakeholders including the test community and users.

12 Final Report: A Final Technical and Business Report shall be prepared at the end of the effort. A final report shall be provided whether any or all of the project Stages are completed. This report takes the place of the last annual report due (a financial report is still required). The report shall narrate a complete summary of the project execution and associated results obtained. The narration will include outstanding problems and their potential solutions, and problems solved during the course of the year(s), along with the solution to the solved problems. The report shall demonstrate how the prototype was advanced.

The offeror shall submit a Draft Final Report by the 30th calendar day following the end of the project period of performance. The Government shall provide comments to the offeror by the 30th calendar day following receipt of the offeror's Draft Final Report. The offeror shall submit the Final Report on the 30th calendar day after receipt of Government comments.

PROTOTYPE DELIVERABLES: To achieve a solution suitable for incorporation into a Program of Record and/or employment in the field, an incremental prototype build-test-fix approach is often beneficial. This iterative approach shall be used incorporating the following activities into the overall project schedule:

1 Initial Prototypes: The awardee(s) will develop up to five (5) initial prototypes to support demonstrations and operational characterization test and evaluation.

2 Final Prototypes: Awardee(s) shall deliver up to ten (10) fully operational prototypes, including associated hardware and software to support test, evaluation, and characterization by 30 Sep 2028 to Aberdeen Proving Ground Edgewood Area (APG-EA).

3 Performance Assessment: The Awardee(s) will propose and execute a performance assessment of the prototypes before delivery to the Government. The awardee(s) will propose performance metrics based on its assessment approach.

4 Test and Integration Support: Planned testing of the CSIRP program will leverage both contractor run testing, independent government testing, and user feedback events. The Awardee(s) will propose appropriate activities to demonstrate product performance, suitability for use in relevant conditions by Warfighters, and technical maturity / supportability as required for insertion into Program(s) of Record or to address Urgent Operational Needs. The following list provides possible considerations for future testing events:

- Sensor performance testing on the detection/measurement accuracy and identification of radiological and nuclear hazards.
- Sensor chemical and/or biological live agent testing (Government executed):
 - Industry partner provides detector technology for Government executed laboratory testing
- Integration/Technical Events:
 - In house integration/testing:
 - Industry partner provides Government documented procedures and results in a test report
 - Government executed technical events:
 - Industry partner trains Government representatives on use of technology, then hands over technology for Government technical testing/demonstration
- User Feedback Events (Government executed):
 - Industry partner provides overview and hands on training / demonstration of technology to collect user feedback in a classroom/simulated operational environment
 - Industry partner supports Government executed Table Top Exercises (TTX) to discuss and document the development of CONOPS and Techniques, Tactics and Procedures (TTPs)
 - Industry partner supports Government organized Soldier Touch Point (STP) to gain Warfighter feedback regarding operation of technology in relevant scenarios
- Operational Demonstration (Government executed):
 - Industry partner supports New Equipment Training (NET) and Doctrine, Tactics, and Techniques (DTT) training to representative users
 - Government sets up realistic scenarios to collect feedback on how users would utilize the technology

For estimating purposes (for the proposed task length), assume the following timeframe(s) and location(s) for Performance Assessments and user feedback events located in Table 1.

Table 1: CSIRP Proposed Test Locations

Potential Event	Potential Location	Estimated Timeframe
Prototype Testing	Eldorado, TX	6-9 months post award
Performance Testing (CB)	Dugway Proving Ground, UT	9-12 months post award
Performance Testing (RN)	Oak Ridge National Labs, TN	9-12 months post award
Agent Testing	APG (Edgewood), MD	12-18 months post award
Agent Testing	Kansas City, MO	12-18 months post award
User Testing	Ft. Leonard Wood, MO	24 months post award
User Testing	Yuma Proving Ground, AZ	30 months post award

PERIOD AND PLACE OF PERFORMANCE: Awardee(s) shall begin performance at award and continue test support through 30 Sep 2028. Prototypes shall be delivered to the Government and training on use of said prototypes shall occur at APG-EA, Gunpowder, MD no later than 30 Sep 2028.

Offerors will choose the test locations and durations from the list in Table 1 in support of government developmental and user testing (June 2026-Sep 2028) as necessary for their intended scope of work. Note: Additional testing locations and durations may be required due to failures or programmatic changes.

GOVERNMENT FURNISHED EQUIPMENT: In support of proposed efforts, the government may be able to provide equipment to performers from the list below.

- Man Transportable Robotic System Increment II (MTRS Inc II)
- SkyRaider R80D unmanned aircraft
- Boston Dynamics SPOT legged robot
- Array Configured of Remote Network Sensors (ACoRNS) payload modules
- Joint Chemical Agent Detector (JCAD)
- Enhanced Chemical Detector Module (eCDM)
- E2MU (RDRoP) Radiation Detector
- Compact Vapor Chemical Agent Detector (CVCAD)
- Proximate Chemical Agent Detector (PCAD) Non-Trace
- MPU-5 Radio
- Silvus Streamcaster Radio
- Platform and sensor data (ICD, TDP, etc.) will be made available to awardees

FUNDING CONFIDENCE LEVEL: CL-1. The Government is highly confident that the funds will be available for this effort.

AGREEMENTS OFFICER REPRESENTATIVE (AOR):

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